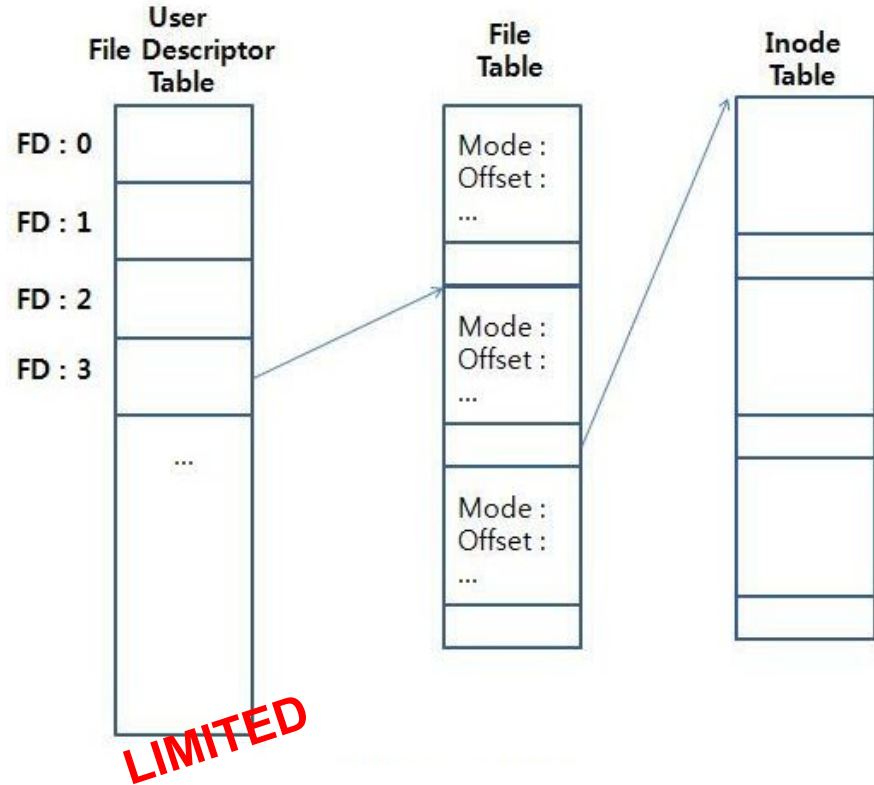


Runtime Resource Leak Repair in Java

노준영
UNIST@LOFT

Background

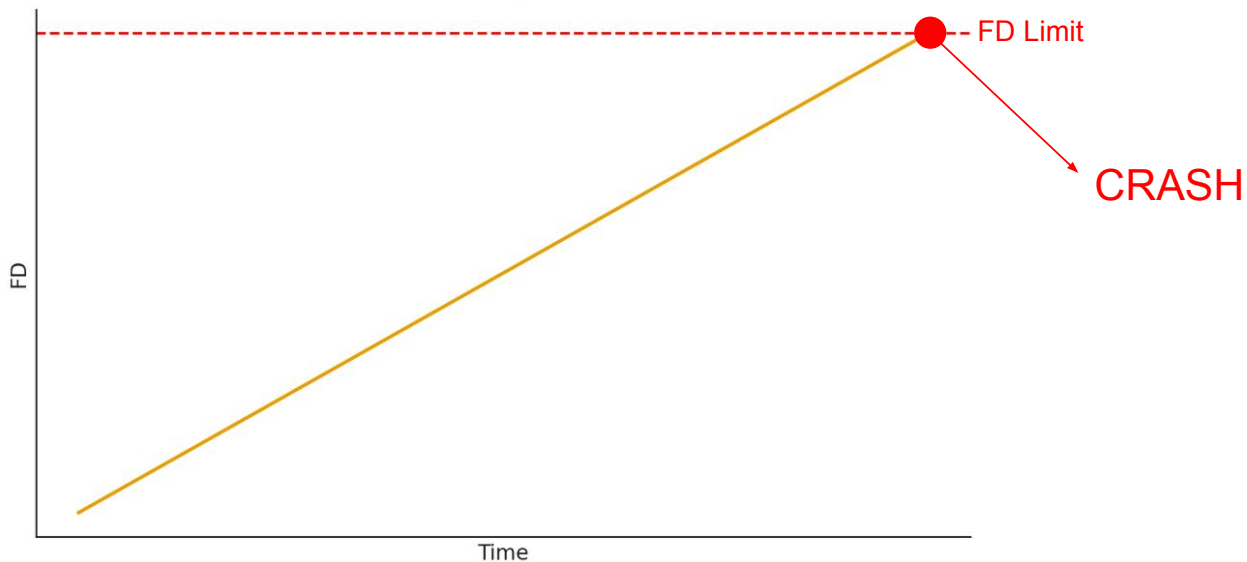
- Memory -> GC (Garbage Collector)
- FileDescriptor (Unix / Linux)
 - > Explicitly close()



FD Resource Leak

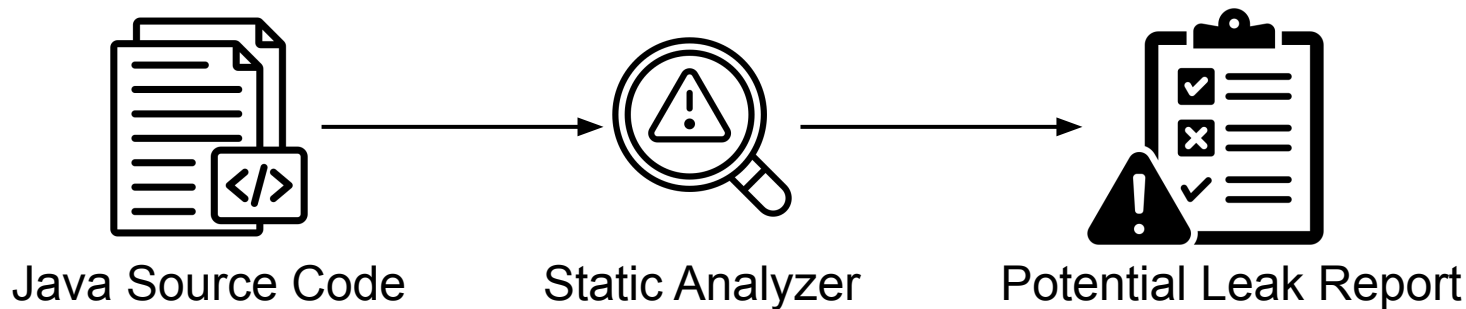
Resource Open
-> FD Allocated

NO CLOSE
-> FD not released(LEAK)



Motivation

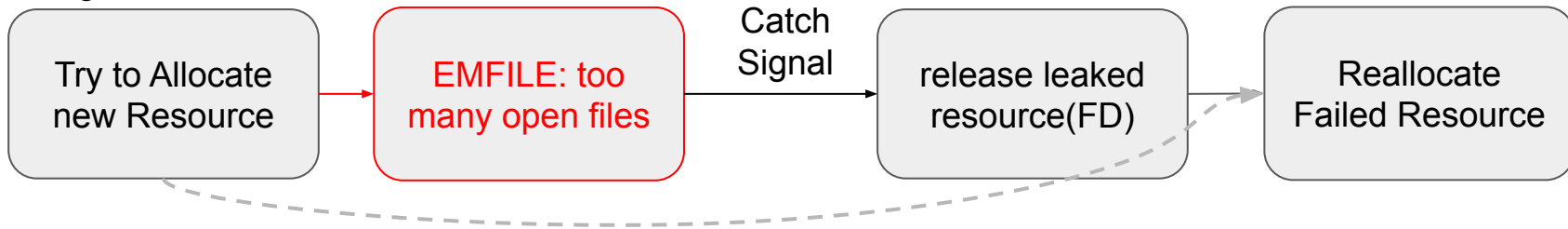
- Previous research -> AOT (Ahead-of-Time) with Static Analyzer



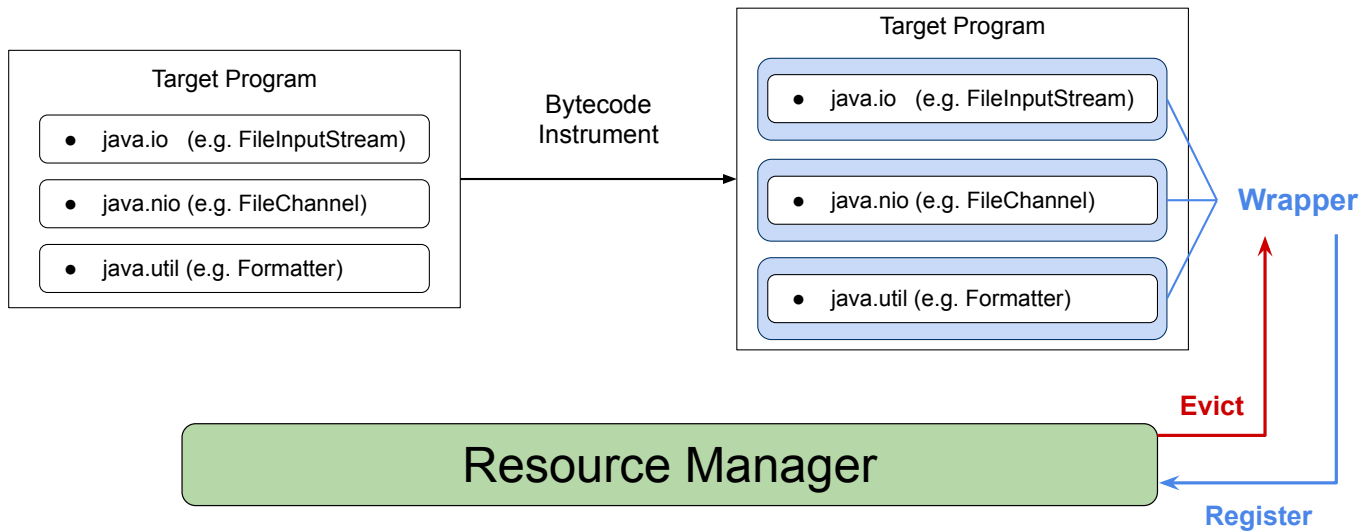
Our Methods

- How about JIT (Just-In-Time)?

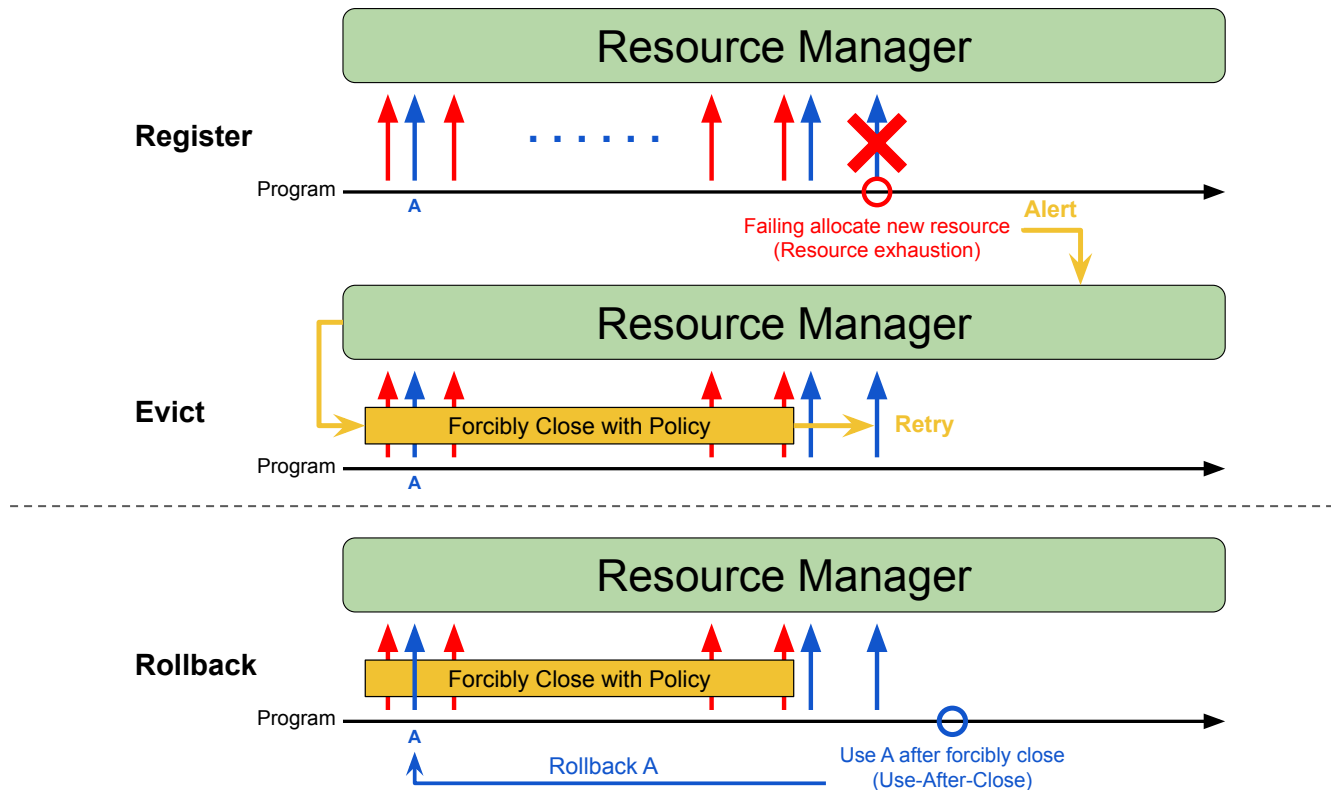
If Program Have Leak



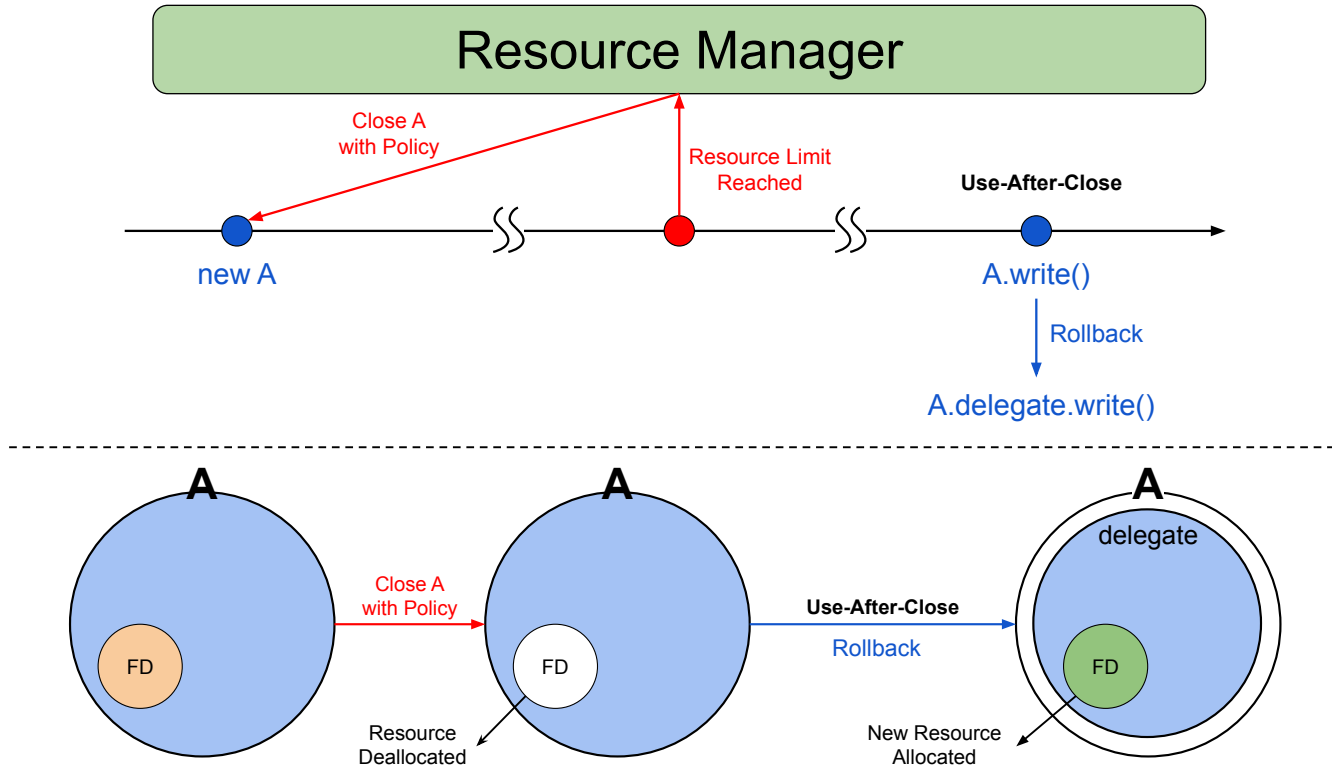
Core Idea



Core Idea



Core Idea - Rollback



Current Progress

- Benchmark Evaluation (Custom crash scenario on NJR-1 Benchmark)
(Crash Prevention, Runtime Overhead)
- Realistic Scenario Evaluation